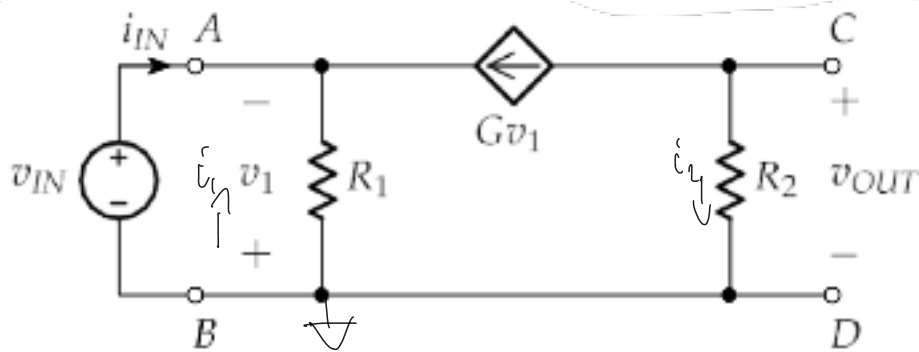




(a)

$$V_1 = -V_{IN} \quad \text{KCL at C: } Gv_1 + i_2 = 0$$

$$\frac{V_{out}}{R_2} = -Gv_1$$

$$V_{out} = -Gv_1 \cdot R_2$$

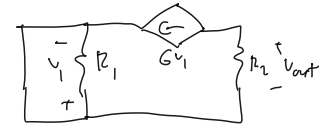
$$V_{out} = -G(-V_{IN}) \cdot R_2$$

$$\frac{V_{out}}{V_{IN}} = GR_2 = A_v \text{ (gain)}$$

for R_{IN} :

KCL at A: $-i_{IN} - Gv_1 - i_1 = 0$
 $i_{IN} = -Gv_1 - \frac{v_1}{R_1}$
 $i_{IN} = -Gv_1 - \frac{(-V_{IN})}{R_1}$
 $i_{IN} = -G(-V_{IN}) + \frac{V_{IN}}{R_1}$
 $i_{IN} = V_{IN} \left(G + \frac{1}{R_1} \right)$
 $R_{IN} = \frac{V_{IN}}{i_{IN}} = \frac{1}{\left(G + \frac{1}{R_1} \right)}$

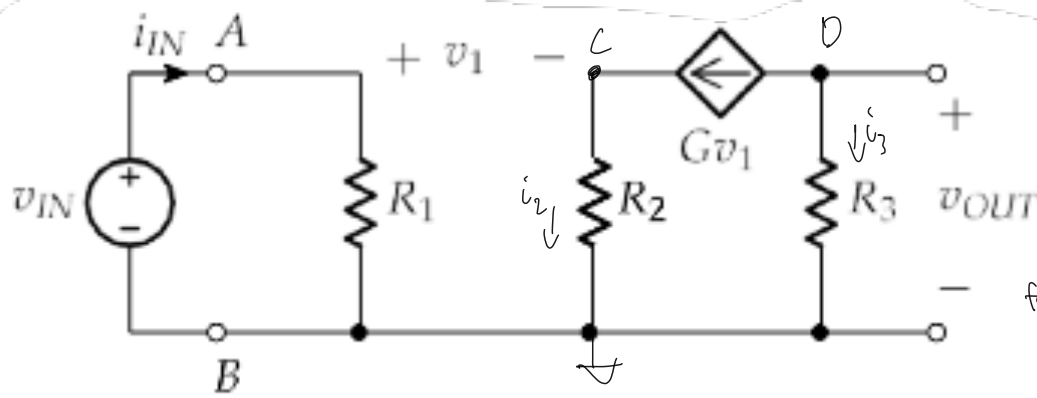
for R_{out} : since V_{out} is across R_2 , $R_{out} = R_2$



$$v_1 = 0$$

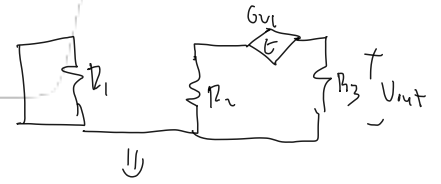
$$\therefore R_2 \left\{ I_{v_{out}} \right.$$

$$R_{out} = R_2$$



for R_{IN} : $R_{IN} = R_1$ since current from v_{IN} flows in loop and doesn't contribute to rest of circuit

for R_{out} : v_{out} defined across R_3 so $R_{out} = R_3$



$$v_1 = -Gv_1 R_2$$

$$v_1(GR_2) = 0 \\ v_1 = 0 \therefore \left\{ R_3 \right\} v_{out} \\ = R_{out}$$

$$\text{KCL @ C: } -Gv_1 + i_2 = 0$$

$$i_2 = Gv_1$$

$$\frac{v_c}{R_2} = Gv_1$$

$$v_c = Gv_1 R_2$$

$$\text{KCL @ D: } Gv_1 + i_3 = 0$$

$$i_3 = -Gv_1$$

$$\frac{v_{out}}{R_3} = -Gv_1$$

$$v_{out} = -Gv_1 R_3$$

$$v_1 = v_{IN} - v_c$$

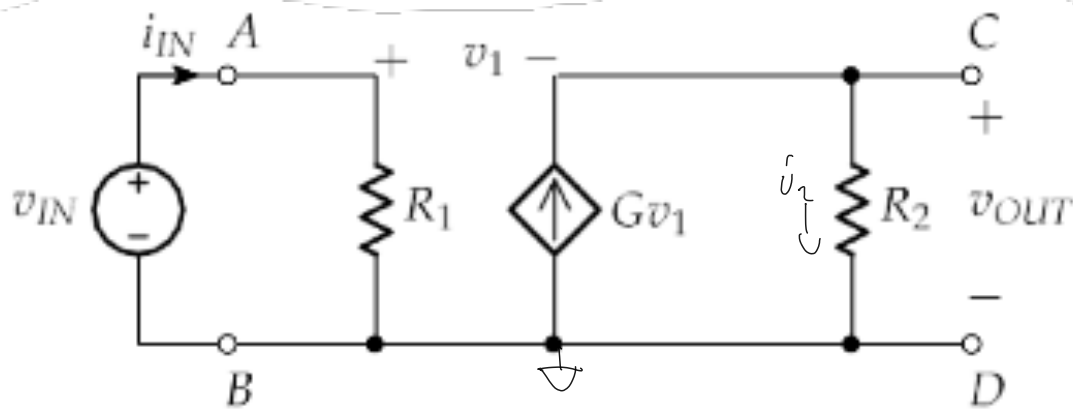
$$v_1 = v_{IN} - Gv_1 R_2$$

$$v_1 + Gv_1 R_2 = v_{IN}$$

$$v_{IN} = v_1 (1 + GR_2)$$

$$A_v = \frac{v_{out}}{v_{IN}} = \frac{-Gv_1 R_3}{v_1 (1 + GR_2)}$$

$$A_v = -\frac{GR_3}{1 + GR_2} \text{ (gain)}$$



$R_1 = R_{IN}$ since i_N flows only in loop
 and $v_{IN} = i_{IN} R_{IN}$
 $\frac{v_{IN}}{i_{IN}} = R_{IN}$
 and i_{IN} only flows thru R_1 $\therefore R_1 = R_{IN}$

(c)

KCL @ C: $-Gv_1 + \frac{v_C}{R_2} = 0$

* $v_C = v_{out}$

$$\frac{v_{out}}{R_2} = Gv_1$$

$$v_1 = v_{IN} - v_C = v_{IN} - v_{out} \quad \frac{v_{out}}{R_2} = G(v_{IN} - v_{out})$$

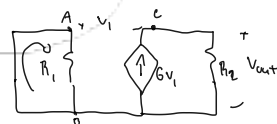
$$\frac{v_{out}}{R_2} = Gv_{IN} - Gv_{out}$$

$$\frac{v_{out}}{R_2} + Gv_{out} = Gv_{IN}$$

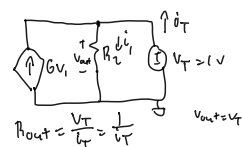
$$v_{out} \left(\frac{1}{R_2} + G \right) = Gv_{IN}$$

$$\frac{v_{out}}{v_{IN}} = \frac{G}{\left(\frac{1}{R_2} + G \right)}$$

for R_{out} :



$v_A = 0$
 $v_C = v_{out}$
 $v_1 = v_A - v_C = 0 - v_C = -v_{out}$
 $v_1 = -v_{out}$



$$R_{out} = \frac{v_T}{i_T} = \frac{1}{i_T}$$

KCL: $-Gv_1 - i_T + i_1 = 0$

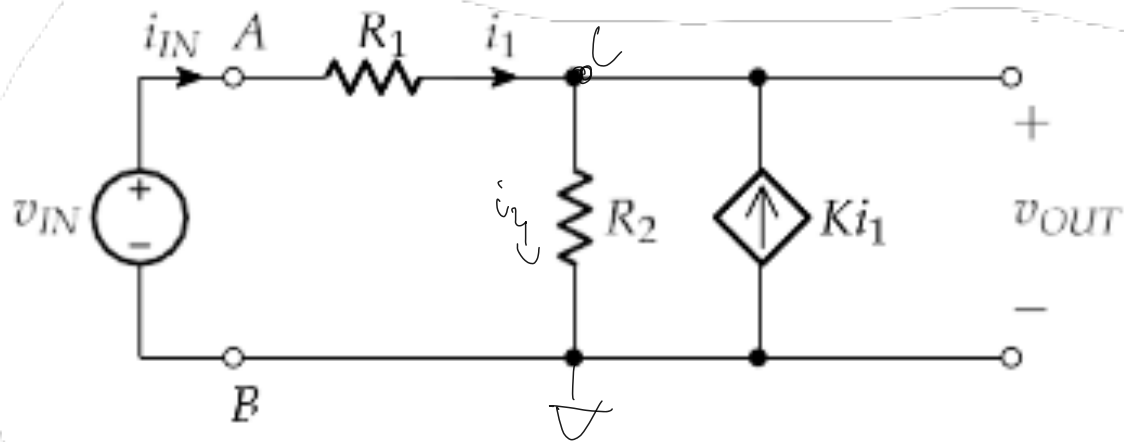
$$i_T = i_1 - Gv_1$$

$$i_T = \frac{v_T}{R_1} - G(-v_{out})$$

$$i_T = \frac{v_T}{R_1} + Gv_T$$

$$i_T = v_T \left(\frac{1}{R_1} + G \right)$$

$$\left(\frac{1}{R_1} + G \right) = \frac{v_T}{i_T} = R_{out}$$



(d)

KCL at V_A : $-i_{IN} + i_1 = 0$

$$i_1 = i_{IN}$$

$$\frac{V_A - V_L}{R_1} = i_{IN}$$

$$\frac{V_{IN} - V_{out}}{R_1} = i_{IN} = i_1$$

KCL at V_L : $-i_1 + i_2 - Ki_1 = 0$

$$-\frac{V_A - V_L}{R_1} + \frac{V_L}{R_2} - K\left(\frac{V_A - V_L}{R_1}\right) = 0$$

$$\frac{V_{out} - V_{IN}}{R_1} + \frac{V_{out}}{R_2} - K\left(\frac{V_{IN} - V_{out}}{R_1}\right) = 0$$

$$\frac{V_{out} - V_{IN}}{R_1} + \frac{V_{out}}{R_2} + K\frac{V_{out} - V_{IN}}{R_1} = 0$$

$$V_{out}\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{K}{R_1}\right) + V_{IN}\left(-\frac{1}{R_1} - \frac{K}{R_1}\right) = 0$$

$$V_{out}\left(\frac{1}{R_2} + \frac{1+K}{R_1}\right) = -V_{IN}\left(-\frac{1+K}{R_1}\right)$$

$$\frac{V_{out}}{V_{IN}} = \frac{(1+K)}{\left(\frac{1}{R_2} + \frac{1+K}{R_1}\right)} = \frac{1+K}{\left(\frac{R_1}{R_2} + 1+K\right)} = A_v \text{ (gain)}$$

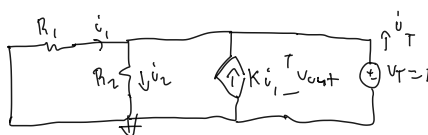
For R_{IN} : $R_{IN} = \frac{V_{IN}}{i_{IN}}$

$$i_{IN} = i_1 = \frac{V_{IN} - V_{out}}{R_1}$$

$$R_{IN} = \frac{V_{IN}}{\frac{V_{IN} - V_{out}}{R_1}}$$

$$R_{IN} = \frac{V_{IN} R_1}{V_{IN} - V_{out}}$$

For R_{out} :



KCL at top node: $-i_1 - Ki_1 - i_T + i_2 = 0$

$$i_T = i_2 - i_1 - Ki_1$$

$$i_T = \frac{V_T}{R_2} - \frac{0 - V_T}{R_1} - K\left(\frac{0 - V_T}{R_1}\right)$$

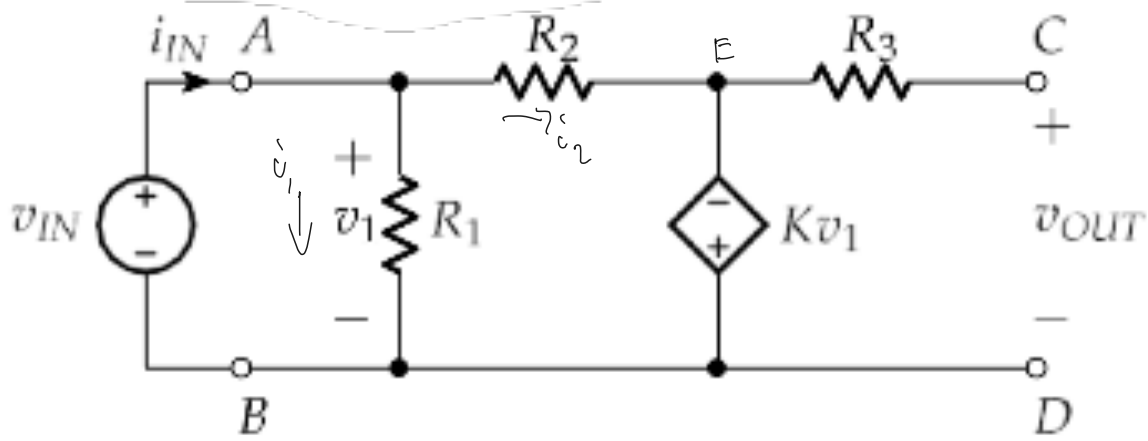
$$i_T = \frac{V_T}{R_2} + \frac{V_T}{R_1} + K\frac{V_T}{R_1}$$

$$i_T = V_T\left(\frac{1}{R_2} + \frac{1}{R_1} + \frac{K}{R_1}\right)$$

$$R_{out} = \frac{V_{out}}{i_{out}} = \frac{V_T}{i_T}$$

$$R_{out} = \frac{V_T}{V_T\left(\frac{1}{R_2} + \frac{1+K}{R_1}\right)}$$

$$R_{out} = \frac{1}{\frac{1}{R_2} + \frac{1+K}{R_1}}$$



(e)

$$V_{IN} = V_1$$

$$V_{out} = -K V_1$$

$$V_{out} = -K V_{IN}$$

$$\frac{V_{out}}{V_{IN}} = -K = A_v \text{ (gain)}$$

$$\text{for } R_{IN} = \frac{V_{IN}}{i_{IN}}$$

$$KCL \propto A: i_{IN} + i_1 + i_2 = 0$$

$$i_{IN} = i_1 + i_2$$

$$i_{IN} = \frac{V_{IN}}{R_1} + \frac{V_{IN} - V_E}{R_2}$$

$$V_E = 0 - K V_1 = -K V_{IN}$$

$$i_{IN} = \frac{V_{IN}}{R_1} + \frac{V_{IN} - (-K V_{IN})}{R_2}$$

$$i_{IN} = V_{IN} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{K}{R_2} \right)$$

$$\frac{V_{IN}}{i_{IN}} = R_{IN} = \frac{1}{\left(\frac{1}{R_1} + \frac{1+K}{R_2} \right)}$$

$$\text{For } R_{out} = \frac{V_{out}}{i_{out}} = \frac{V_T}{i_T} = \frac{1}{i_T}$$

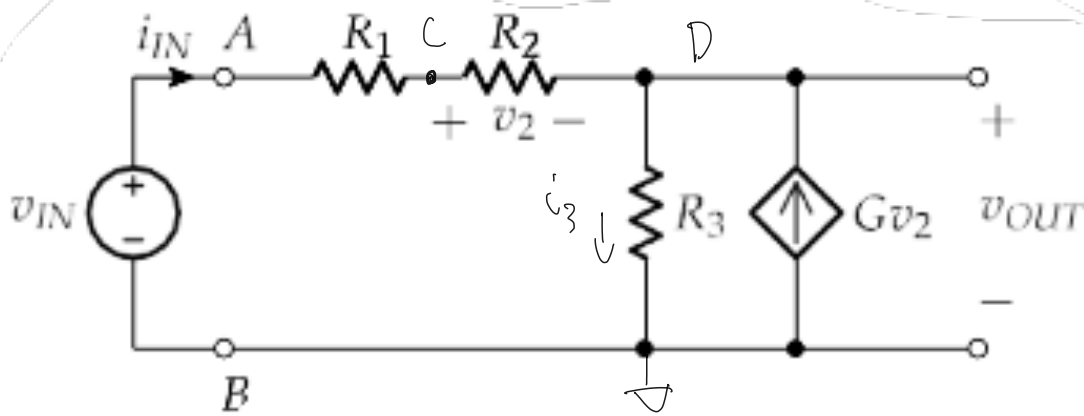


short: $v_1 = 0 \therefore K v_1 = 0$



short: $\frac{V_T}{i_T} = R_{out}$

$$\frac{V_T}{i_T} = R_{out} = R_2$$



$$v_2 = i_{IN} R_2$$

$$= \left(\frac{V_{IN} - V_{out}}{R_1 + R_2} \right) R_2 \quad (f)$$

$$V_D = V_{out}$$

KCL @ D: $-i_N + i_3 - Gv_2 = 0$

$$-\left(\frac{V_{IN} - V_D}{R_1 + R_2} \right) + \frac{V_D}{R_3} - Gv_2 = 0$$

$$\frac{V_{out} - V_{IN}}{R_1 + R_2} + \frac{V_{out}}{R_3} - G \left(\left(\frac{V_{IN} - V_{out}}{R_1 + R_2} \right) R_2 \right) = 0$$

$$V_{out} \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} + \frac{GR_2}{R_1 + R_2} \right) + V_{IN} \left(-\frac{1}{R_1 + R_2} - \frac{GR_2}{R_1 + R_2} \right) = 0$$

$$V_{out} \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} + \frac{GR_2}{R_1 + R_2} \right) = -V_{IN} \left(-\frac{1}{R_1 + R_2} - \frac{GR_2}{R_1 + R_2} \right)$$

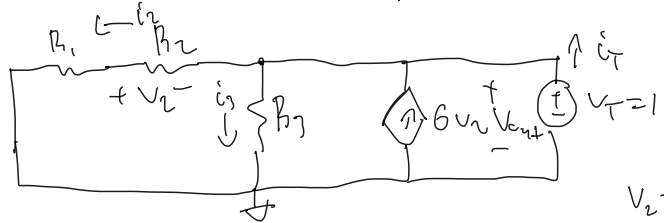
$$\frac{V_{out}}{V_{IN}} = \frac{\left(\frac{1 + GR_2}{R_1 + R_2} \right)}{\frac{1 + GR_2}{R_1 + R_2} + \frac{1}{R_3}} = \frac{1 + GR_2}{1 + GR_2 + \frac{R_1 + R_2}{R_3}} = A_v \text{ (gain)}$$

For $R_{IN} = \frac{V_{IN}}{i_{IN}}$

$$i_{IN} = \frac{V_{IN} - V_{out}}{R_1 + R_2}$$

$$R_{IN} = \frac{V_{IN}}{\frac{V_{IN} - V_{out}}{R_1 + R_2}} = \frac{V_{IN} (R_1 + R_2)}{V_{IN} - V_{out}}$$

For $R_{out} = \frac{V_{out}}{i_{out}} = \frac{V_T}{i_T}$



$$v_2 = -i_2 R_2$$

$$v_2 = -\left(\frac{V_T}{R_1 + R_2} \right) R_2$$

KCL @ top: $-i_T - Gv_2 + i_3 + i_2 = 0$

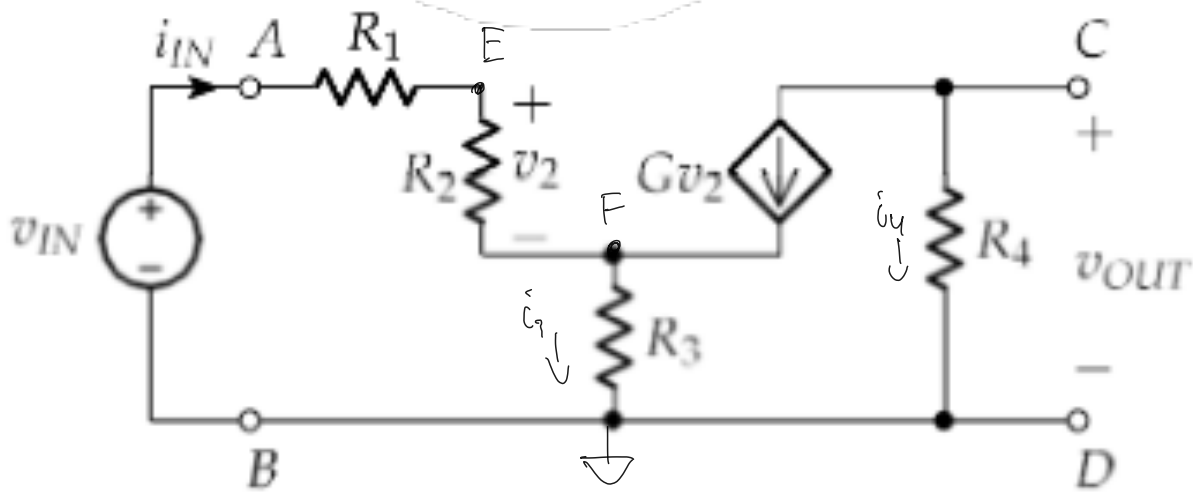
$$-i_T - Gv_2 + \frac{V_T}{R_3} + \frac{V_T}{R_1 + R_2} = 0$$

$$-i_T - G \left(-\left(\frac{V_T}{R_1 + R_2} \right) R_2 \right) + \frac{V_T}{R_3} + \frac{V_T}{R_1 + R_2} = 0$$

$$i_T = V_T \left(\frac{GR_2}{R_1 + R_2} + \frac{1}{R_3} + \frac{1}{R_1 + R_2} \right)$$

$$i_T = V_T \left(\frac{GR_2 + 1}{R_1 + R_2} + \frac{1}{R_3} \right)$$

$$\frac{V_T}{i_T} = R_{out} = \frac{1}{\frac{GR_2 + 1}{R_1 + R_2} + \frac{1}{R_3}}$$



(g)

$$\frac{V_{IN} - V_F}{R_1 + R_2} = i_{IN} \quad V_2 = V_E - V_F = i_N R_2 = \frac{V_{IN} - V_F}{R_1 + R_2} R_2$$

KCL @ F: $-\dot{i}_{IN} + \dot{i}_3 - Gv_2 = 0$

$$-\left(\frac{V_{IN} - V_F}{R_1 + R_2}\right) + \frac{V_F}{R_3} - Gv_2 = 0$$

$$\frac{V_F - V_{IN}}{R_1 + R_2} + \frac{V_F}{R_3} - G\left(\frac{V_{IN} - V_F}{R_1 + R_2}\right) R_2 = 0$$

$$V_F \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} + \frac{GR_2}{R_1 + R_2} \right) + V_{IN} \left(\frac{-1}{R_1 + R_2} - \frac{GR_2}{R_1 + R_2} \right) = 0$$

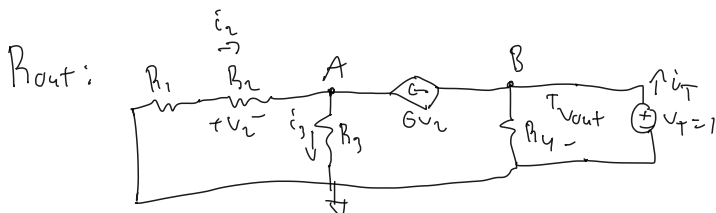
$$V_F = V_{IN} \frac{\left(\frac{1 + GR_2}{R_1 + R_2} \right)}{\frac{1 + GR_2}{R_1 + R_2} + \frac{1}{R_3}}$$

$$V_F = \frac{1 + GR_2}{1 + GR_2 + \frac{R_1 + R_2}{R_3}}$$

$$V_F = \left(\frac{R_3 (1 + GR_2)}{R_3 (1 + GR_2) + R_1 + R_2} \right) V_{IN} \Rightarrow$$

For $R_{IN} = \frac{V_{IN}}{\dot{i}_{IN}} = \frac{V_{IN}}{\frac{V_2}{R_2}} = \frac{V_{IN}}{\frac{V_{IN}}{D}} = D$

$$R_{IN} = R_3 (1 + GR_2) + R_1 + R_2$$



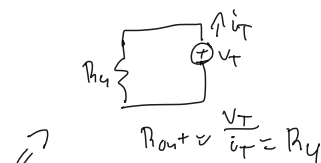
KCL @ A: $-\dot{i}_2 + \dot{i}_3 - Gv_2 = 0$

$$-\left(\frac{0 - V_A}{R_1 + R_2}\right) + \frac{V_A}{R_3} - Gv_2 = 0$$

$$\frac{V_A}{R_1 + R_2} + \frac{V_A}{R_3} - G\left(R_2 \left(\frac{-V_A}{R_1 + R_2}\right)\right) = 0$$

$$V_A \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3} + \frac{GR_2}{R_1 + R_2} \right) = 0$$

$V_A = 0 \therefore V_A = 0$ on $V_2 = 0 - V_A = 0 - 0 \therefore V_2 = 0$



$$R_{out} = \frac{v_T}{i_T} = R_4$$

KCL @ C: $\dot{i}_4 + Gv_2 = 0$

$$\frac{V_{out}}{R_4} + Gv_2 = 0$$

$$\frac{V_{out}}{R_4} = -G\left(\frac{R_2}{D} V_{IN}\right)$$

$$\frac{V_{out}}{V_{IN}} = -\frac{R_2 R_4}{D} G$$

$$A_v (\text{gain}) = -G \frac{R_2 R_4}{R_3 (1 + GR_2) + R_1 + R_2}$$

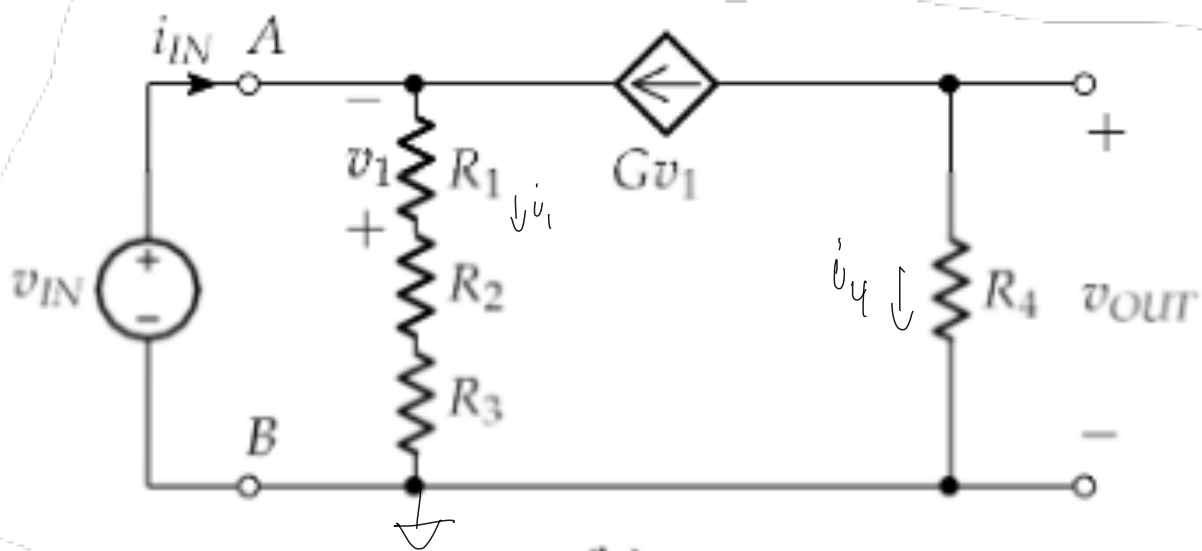
$$V_2 = \frac{R_2}{R_1 + R_2} \left(V_{IN} - \left[V_{IN} \left(\frac{R_3 (1 + GR_2)}{R_3 (1 + GR_2) + R_1 + R_2} \right) \right] \right) \quad D = R_3 (1 + GR_2) + R_1 + R_2$$

$$V_2 = \frac{R_2}{R_1 + R_2} \left[V_{IN} \left(1 - \frac{R_3 (1 + GR_2)}{D} \right) \right]$$

$$= \frac{R_2}{R_1 + R_2} \left[V_{IN} \left(\frac{D - R_3 (1 + GR_2)}{D} \right) \right]$$

$$V_2 = \frac{R_2}{R_1 + R_2} \left[V_{IN} \left(\frac{R_3 (1 + GR_2) + R_1 + R_2 - R_3 (1 + GR_2)}{D} \right) \right]$$

$$V_2 = R_2 \frac{V_{IN}}{D}$$



(h)

$$v_1 = -i_1 R_1 = -\left(\frac{v_{IN}}{R_1 + R_2 + R_3}\right) R_1$$

KCL @ C: $Gv_1 + i_4 = 0$

$$G\left(-\frac{v_{IN}}{R_1 + R_2 + R_3}\right) R_1 + \frac{v_{out}}{R_4} = 0$$

$$v_{IN} \left(-\frac{GR_1}{R_1 + R_2 + R_3}\right) = -\frac{v_{out}}{R_4}$$

$$\frac{v_{out}}{v_{IN}} = \frac{GR_1 R_4}{R_1 + R_2 + R_3} = A_v (\text{gain})$$

For $R_{IN} = \frac{v_{IN}}{i_{IN}}$

KCL @ A: $-i_{IN} + i_1 - Gv_1 = 0$

$$\frac{v_{IN}}{R_1 + R_2 + R_3} = i_{IN} + G\left(-\frac{v_{IN}}{R_1 + R_2 + R_3}\right) R_1$$

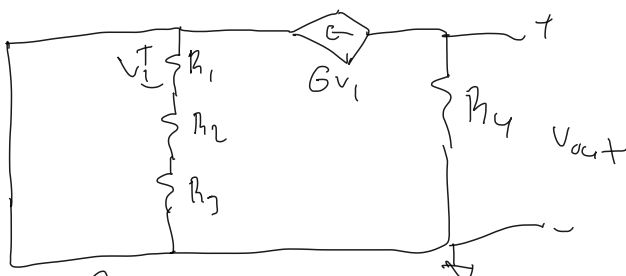
$$v_{IN} \left(\frac{1}{R_1 + R_2 + R_3} + \frac{GR_1}{R_1 + R_2 + R_3}\right) = i_{IN}$$

$$i_{IN} = v_{IN} \left(\frac{1 + GR_1}{R_1 + R_2 + R_3}\right)$$

so $R_{IN} = \frac{v_{IN}}{i_{IN}} = \frac{v_{IN}}{v_{IN} \left(\frac{1 + GR_1}{R_1 + R_2 + R_3}\right)}$

$$R_{IN} = \frac{R_1 + R_2 + R_3}{1 + GR_1}$$

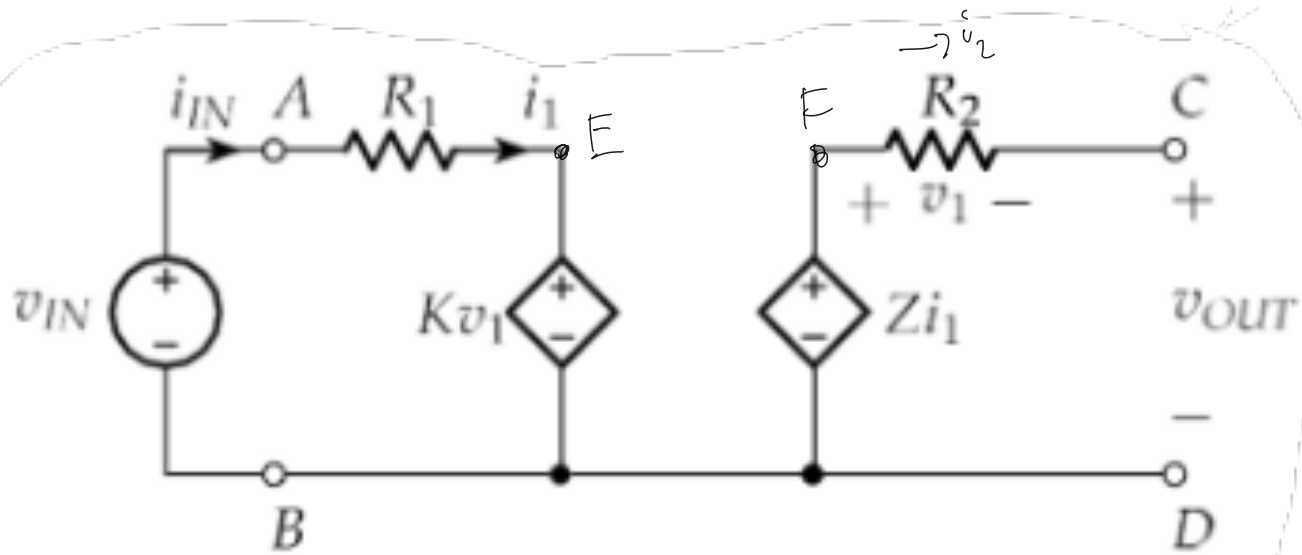
For R_{out}



↑
short $\Rightarrow$
 $v_1 = 0$



$\therefore R_{out} = R_4$



(i)

$$i_N = i_1 = \frac{V_{IN} - Kv_1}{R_1}$$

$$i_1 = \frac{V_{IN} - 0}{R_1} = i_{IN}$$

$$v_1 = Zi_1 - v_{out}$$

$$i_2 = 0 = \frac{Zi_1 - v_{out}}{R_2}$$

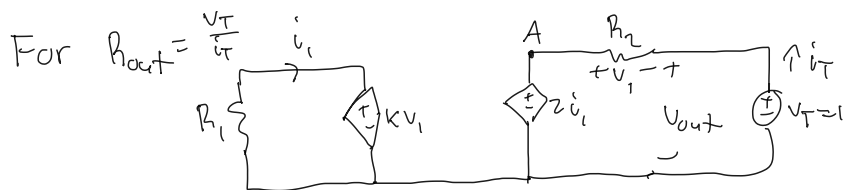
$$Zi_1 = v_{out} \therefore v_1 = 0$$

$$Z\left(\frac{V_{IN}}{R_1}\right) = v_{out}$$

$$\frac{v_{out}}{V_{IN}} = \frac{Z}{R_1} = A_V (\text{gain})$$

$$\text{For } R_{IN} = \frac{v_{IN}}{i_{IN}} = \frac{v_{IN}}{\frac{v_{IN}}{R_1}} = R_1$$

$$R_{IN} = R_1$$



$$\frac{v_T - v_A}{R_2} = i_T \quad i_1 = -\left(\frac{Kv_1}{R_1}\right) \quad v_1 = v_A - v_T \quad v_A = Zi_1 = -\frac{KZ}{R_1} v_1$$

$$\frac{v_T - \frac{KZ}{R_1} (v_A - v_T)}{R_2} = i_T$$

$$\frac{R_1 + KZ - KZ}{R_1 + KZ} = i_T$$

$$\frac{R_1}{R_2(R_1 + KZ)} = i_T$$

$$R_{out} = \frac{v_T}{i_T} = \frac{1}{\frac{R_1}{R_2(R_1 + KZ)}}$$

$$R_{out} = \frac{R_2(R_1 + KZ)}{R_1}$$

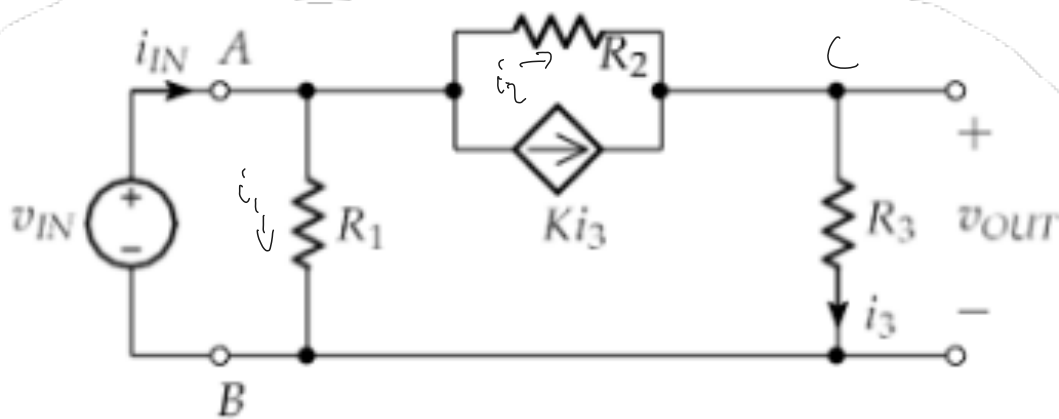
$$v_A = -\frac{KZ}{R_1} (v_A - v_T)$$

$$v_A \left(1 + \frac{KZ}{R_1}\right) = v_T \frac{KZ}{R_1}$$

$$v_A = v_T \frac{\frac{KZ}{R_1}}{1 + \frac{KZ}{R_1}}$$

$$v_A = v_T \frac{KZ}{R_1 + KZ}$$

$$v_A = \frac{KZ}{R_1 + KZ}$$



(i)

KCL @ C: $-i_2 - K i_3 + i_3 = 0$

$$-\left(\frac{V_{IN} - V_{OUT}}{R_2}\right) - K\left(\frac{V_{OUT}}{R_3}\right) + \frac{V_{OUT}}{R_3} = 0$$

$$V_{OUT} \left(\frac{1}{R_2} - \frac{K}{R_3} + \frac{1}{R_3} \right) + V_{IN} \left(-\frac{1}{R_2} \right) = 0$$

$$V_{OUT} \left(\frac{1}{R_2} + \frac{1-K}{R_3} \right) = V_{IN} \cdot \frac{1}{R_2}$$

$$\frac{V_{OUT}}{V_{IN}} = \frac{1}{R_2} \cdot \frac{1}{\frac{1}{R_2} + \frac{1-K}{R_3}}$$

$$= \frac{1}{1 + \frac{R_2(1-K)}{R_3}}$$

$$= \frac{R_3}{R_3 + R_2(1-K)} = A_V (\text{gain})$$

For $R_{IN} = \frac{V_{IN}}{i_{IN}}$

from KCL @ C: $\frac{V_{OUT}}{R_3} = \frac{V_{IN} - V_{OUT}}{R_2} + K \left(\frac{V_{OUT}}{R_3} \right)$

$V_{OUT} = A_V V_{IN}$

$$\frac{A_V V_{IN}}{R_3} = \frac{V_{IN} - A_V V_{IN}}{R_2} + A_V V_{IN} \frac{K}{R_3}$$

$$\frac{A_V}{R_3} = \frac{1 - A_V}{R_2} + A_V \frac{K}{R_3}$$

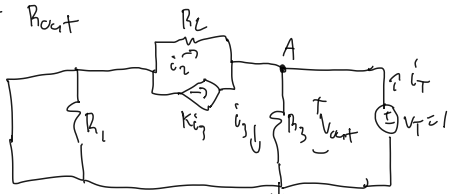
$$\frac{1 - A_V}{R_2} = A_V \left(\frac{1-K}{R_3} \right)$$

$$A_V \left(\frac{1}{R_2} + \frac{1-K}{R_3} \right) = \frac{1}{R_2}$$

$$A_V \left(\frac{R_3 + R_2(1-K)}{R_2 R_3} \right) = \frac{1}{R_2}$$

$$A_V = \left(\frac{R_3}{R_3 + R_2(1-K)} \right)$$

For R_{OUT}



KCL @ A: $-i_T + i_3 - K i_3 - i_2 = 0$

$$i_T = i_3(1-K) - i_2$$

$$i_T = \frac{v_T}{R_3}(1-K) - \frac{0 - v_T}{R_2}$$

$$i_T = \frac{1-K}{R_3} + \frac{1}{R_2} = \frac{R_2(1-K) + R_3}{R_2 R_3}$$

$$R_{OUT} = \frac{v_T}{i_T} = \frac{1}{\frac{R_2(1-K) + R_3}{R_2 R_3}}$$

$$R_{OUT} = \frac{R_2 R_3}{R_2(1-K) + R_3}$$

KCL @ A: $-i_{IN} + i_1 + i_2 + K i_3 = 0$

$$i_{IN} = \frac{V_{IN}}{R_1} + \frac{V_{IN} - V_{OUT}}{R_2} + K \left(\frac{V_{OUT}}{R_3} \right)$$

$$i_{IN} = \frac{V_{IN}}{R_1} + \frac{V_{IN} - A_V V_{IN}}{R_2} + K \frac{A_V V_{IN}}{R_3}$$

$$i_{IN} = V_{IN} \left(\frac{1}{R_1} + \frac{1 - A_V}{R_2} + \frac{K A_V}{R_3} \right)$$

$$= V_{IN} \left(\frac{1}{R_1} + \frac{A_V(1-K) + K A_V}{R_3} \right)$$

$$i_{IN} = V_{IN} \left(\frac{1}{R_1} + \frac{A_V}{R_3} \right)$$

$$i_{IN} = V_{IN} \left(\frac{1}{R_1} + \frac{R_3}{R_3(R_3 + R_2(1-K))} \right)$$

$$i_{IN} = V_{IN} \left(\frac{1}{R_1} + \frac{1}{R_3 + R_2(1-K)} \right)$$

$$\frac{V_{IN}}{i_{IN}} = R_{IN} = \frac{1}{\left(\frac{1}{R_1} + \frac{1}{R_3 + R_2(1-K)} \right)}$$